

Digital Asset Cryptography

Reading Guide

Compiled by the Spaced Research Ledger

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Digital Asset Cryptography is a survey of a threat that has not arrived and of an industry deciding what to do about it anyway. No quantum computer capable of breaking a deployed cryptographic key exists. Several million bitcoin would be exposed if one did. Between those two facts sits everything in this collection: contested demonstrations, rival hardware estimates, draft protocol changes, custody products and the beginnings of regulatory pressure.

The collection covers the subject in five parts. Part 1 covers the threat itself, what it would take to break the cryptography and who claims to know. Part 2 covers how much bitcoin is exposed, a question whose answers differ by millions of coins for definitional reasons. Parts 3 and 4 cover the two big protocol responses, Bitcoin's contested drafts and Ethereum's staffed roadmap. Part 5 covers the custodians who cannot wait for either.

Read front to back, the parts build on each other. The exposure numbers in Part 2 make more sense once Part 1 has explained what an attacker would actually need. The custody work in Part 5 is a response to the protocol stalemate in Part 3. Read alone, every part stands on its own.

Every statement carries a numbered citation to a public source, listed at the end of its part. When sources disagree, the disagreement is reported under its own heading rather than resolved by guesswork. Two habits of this subject deserve a warning up front. Vendors selling quantum urgency are heavily represented in the source pool, and are identified as vendors where they appear. And no exposure figure means anything without its definition, a point Part 2 makes at length.

How to Read This Collection

Each part opens with a summary written for someone who reads nothing else. Numbered sections follow, and each section starts with a short italic paragraph explaining what the thing is and why it matters. Inside each section, Recent Developments covers what changed lately and Current State covers what is durably true. Unresolved Conflicts, where present, records disagreements between sources with a note on what would settle each one.

Technical terms are explained once, on first appearance, and then used normally. Numbers are given in the form the sources used them.

The Parts

Part 1 - ECC Attack Demonstrations & Resource Estimates. What it would take to break the cryptography, according to the people trying to find out. A prize-winning demonstration that Bitcoin developers took apart, and two serious estimates of the required hardware that disagree by a factor of fifty.

Part 2 - Exposure & Migration Analytics. Between 1.6 million and 7 million bitcoin are exposed, depending on the definition. The analytics firms agree on the blockchain data and disagree about what to count, and this part carries the estimates as a series with the dispute attached.

Part 3 - Protocol Proposals & Activation Politics. Bitcoin's draft answers: an address type with no exposed key path, a phased sunset of legacy signatures, and the politics that decide whether anything activates. Includes the workaround that reached mainnet without anyone's permission, at two hundred dollars a transaction.

Part 4 - Protocol Quantum Roadmap. Ethereum's answer, the only staffed and scheduled migration programme among the major chains. The hard problem is compressing half a million validator signatures into one, and most of the engineering exists to solve exactly that.

Part 5 - Institutional Custody & Exchanges. What the companies holding other people's coins have actually shipped, promised and disclosed. Around them, task forces, questionnaires and insurance underwriters have started asking for dates.

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.